

Linear algebra review

- ▶ norm, inner product
- ▶ subspaces, halfspaces
- ▶ independence, basis, dimension
- ▶ invertible matrices

Norms, inner products

- ▶ the inner product

$$x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

also called the *dot product* or *scalar product*

- ▶ the Euclidean norm measures length of vector from origin

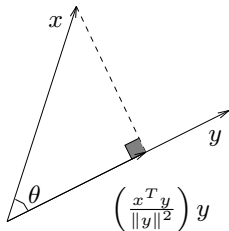
$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} = \sqrt{x^T x}$$

- ▶ the Cauchy-Schwarz inequality

$$|x^T y| \leq \|x\| \|y\|$$

- ▶ the (unsigned) angle between vectors in $\mathbb{R}^n$ is

$$\theta = \angle(x, y) = \cos^{-1} \frac{x^T y}{\|x\| \|y\|}$$



Linear functionals

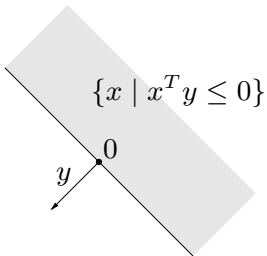
- ▶ a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called a *functional*
- ▶ for $a \in \mathbb{R}^n$, the function $f(x) = a^\top x$ is a *linear functional*
- ▶ every linear functional has this form
- ▶ if $a \neq 0$ then for any $\alpha \in \mathbb{R}$ the set $H_\alpha = \{x \in \mathbb{R}^n \mid a^\top x = \alpha\}$ is called a *hyperplane*
- ▶ hyperplanes H_α and H_β are parallel, H_0 passes through the origin

Halfspaces

- ▶ define a *halfspace* with outward normal vector y , and boundary passing through 0 by

$$H = \{x \mid x^T y \leq 0\}$$

- ▶ H is all vectors x that make an obtuse or right angle with y



Subspaces

A set $S \subset \mathbb{R}^n$ is called a subspace if

$$\begin{array}{ll} x + y \in S & \text{for all } x, y \in S \\ \alpha x \in S & \text{for all } \alpha \in \mathbb{R} \text{ and } x \in S \end{array}$$

- ▶ we say S is *closed under addition* and *closed under scalar multiplication*
- ▶ geometrically, S is a flat set which passes through the origin

Examples of subspaces

- ▶ $S_1 = \mathbb{R}^n$, i.e., the entire vector space is considered a subspace of itself
- ▶ $S_2 = \{0\}$, the origin is the smallest subspace of $\mathbb{R}^n$
- ▶ the *span* of a set of vectors $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ is a subspace

$$\text{span}(v_1, v_2, \dots, v_k) = \{\alpha_1 v_1 + \dots + \alpha_k v_k \mid \alpha_i \in \mathbb{R}\}$$

the set of all *linear combinations* of the vectors

- ▶ the *sum of two subspaces* is a subspace

$$S + T = \{x + y \mid x \in S, y \in T\}$$

Orthogonal subspaces

- ▶ two subspaces $S, T \subset \mathbb{R}^n$ are called *orthogonal* if

$$x^T y = 0 \quad \text{for all } x \in S, y \in T$$

- ▶ for any set $S \subset \mathbb{R}^n$, the *orthogonal complement* is

$$S^\perp = \{x \mid x^T y = 0 \text{ for all } y \in S\}$$

- ▶ $S^\perp$ is always a subspace, even if S is not
- ▶ $S^\perp$ is the set of all vectors x , each of which is orthogonal to every vector in S

Independent set of vectors

a set of vectors $\{v_1, v_2, \dots, v_k\}$ is *independent* if

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0 \quad \implies \quad \alpha_1 = \alpha_2 = \dots = 0$$

some equivalent conditions:

- coefficients of $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k$ are uniquely determined, *i.e.*,

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_k v_k$$

implies $\alpha_1 = \beta_1, \alpha_2 = \beta_2, \dots, \alpha_k = \beta_k$

- no vector v_i can be expressed as a linear combination of the other vectors $v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_k$
- no one vector v_i is in the span of the others

Basis and dimension

set of vectors $\{v_1, v_2, \dots, v_k\}$ is called a *basis* for a subspace S if

$$\begin{aligned} S &= \text{span}(v_1, v_2, \dots, v_k) \\ &\text{and} \\ \{v_1, v_2, \dots, v_k\} &\text{ is independent} \end{aligned}$$

► equivalently, every $v \in S$ *can be uniquely* expressed as

$$v = \alpha_1 v_1 + \dots + \alpha_k v_k$$

► for a given subspace S , the number of vectors in any basis is the same, called the *dimension* of S , denoted $d = \mathbf{dim} S$

► any set of independent vectors in S has no more than d elements

► any set of vectors that span S has at least d elements

Invertibility

A *square* matrix $A \in \mathbb{R}^{n \times n}$ is called *invertible* if there is a matrix $B \in \mathbb{R}^{n \times n}$ such that

$$BA = I$$

- ▶ B is called the *inverse* of A , written A^{-1}
- ▶ we have $AA^{-1} = A^{-1}A = I$
- ▶ A is invertible iff it has linearly independent columns

Interpretations of inverse

suppose $A \in \mathbb{R}^{n \times n}$ has inverse $B = A^{-1}$

- ▶ mapping associated with B undoes mapping associated with A (applied either before or after!)
- ▶ $x = By$ is a perfect (pre- or post-) *equalizer* for the *channel* $y = Ax$
- ▶ $x = By$ is unique solution of $Ax = y$

Change of coordinates

- ▶ standard basis vectors in $\mathbb{R}^n$ are $e_1, e_2, \dots, e_n$, where $e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ has a 1 in i th component

- ▶ for any x we have

$$x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n$$

where x_i are called the *coordinates* of x (in the standard basis)

- ▶ if $t_1, t_2, \dots, t_n$ is another basis for $\mathbb{R}^n$, we have

$$x = \tilde{x}_1 t_1 + \tilde{x}_2 t_2 + \dots + \tilde{x}_n t_n$$

where $\tilde{x}_i$ are the coordinates of x in the basis $t_1, t_2, \dots, t_n$

- ▶ then $x = T\tilde{x}$ and $\tilde{x} = T^{-1}x$

Similarity transformation

consider linear transformation $y = Ax$, $A \in \mathbb{R}^{n \times n}$

express y and x in terms of $t_1, t_2, \dots, t_n$, so $x = T\tilde{x}$ and $y = T\tilde{y}$, then

$$\tilde{y} = (T^{-1}AT)\tilde{x}$$

- ▶ $A \longrightarrow T^{-1}AT$ is called *similarity transformation*
- ▶ similarity transformation by T expresses linear transformation $y = Ax$ in coordinates $t_1, t_2, \dots, t_n$